Credit Risk Prediction using Machine Learning

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Abstract

Credit risk assessment is a fundamental task in financial systems where incorrect decisions may lead to significant financial loss. This paper presents an end-to-end machine learning pipeline for classifying credit applicants as "good" or "bad" risk. Logistic Regression and Random Forest models are evaluated, and feature importance is analyzed. The results highlight the effectiveness of ensemble methods for structured financial data.

Keywords: Credit Risk, Machine Learning, Random Forest, Logistic Regression, Classification

1 Introduction

Credit risk prediction is essential in financial institutions. Machine learning methods provide a data-driven alternative to traditional scoring systems.

This study aims to:

- Develop a classification pipeline
- Compare linear and nonlinear models
- Analyze feature importance

2 Dataset

The dataset contains 1000 samples and 20 features from the OpenML German Credit dataset [5]. The target variable is binary:

- Good credit: 700 samples
- Bad credit: 300 samples

3 Methodology

3.1 Preprocessing

We use a `ColumnTransformer`:

- Numerical: median imputation + scaling
- Categorical: mode imputation + one-hot encoding

3.2 Models

Logistic Regression [3]: Linear baseline model with interpretability.

Random Forest [2]: Ensemble model capturing nonlinear relationships.

4 Experiments and Results

4.1 ROC Curve

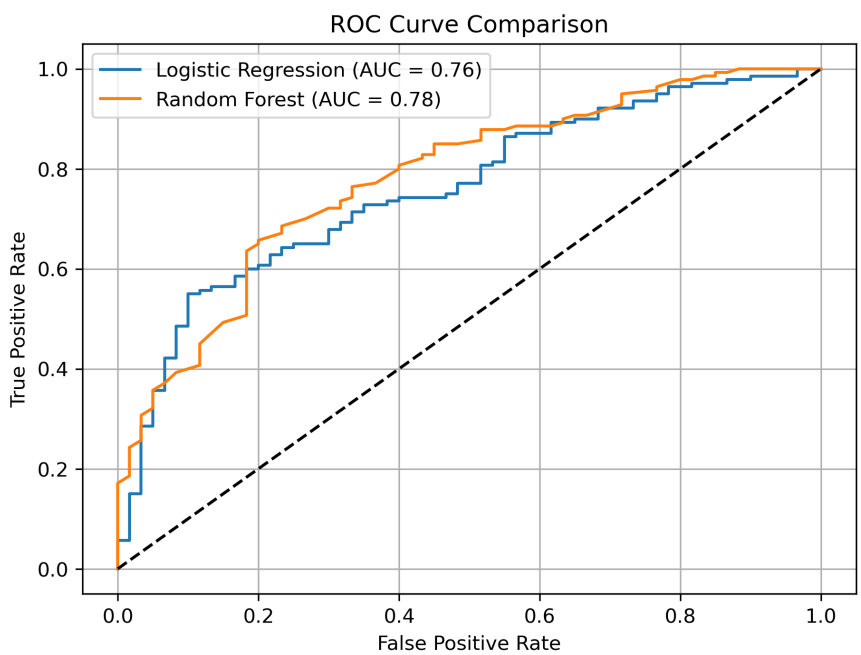

Figure 1: ROC comparison of Logistic Regression and Random Forest

4.2 Confusion Matrix

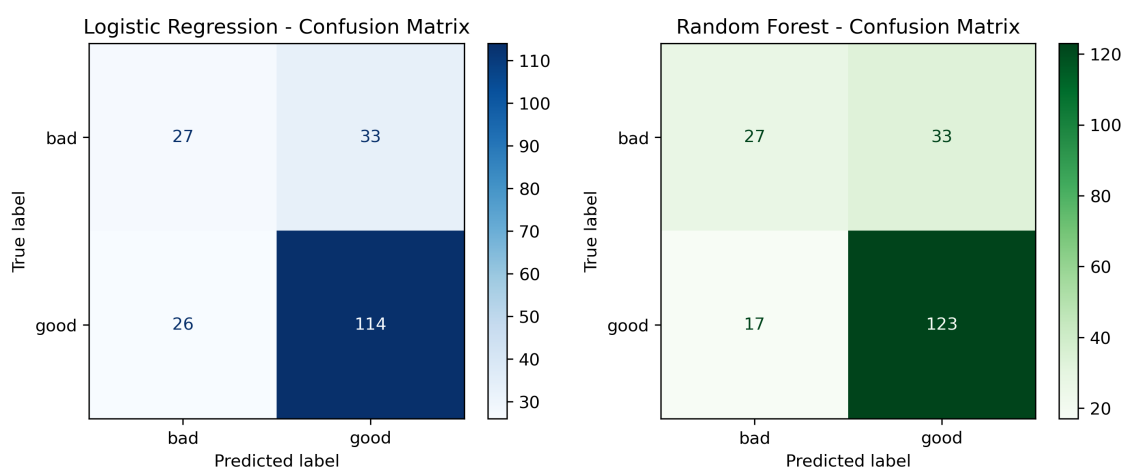

Figure 2: Confusion matrices for both models

4.3 Feature Importance

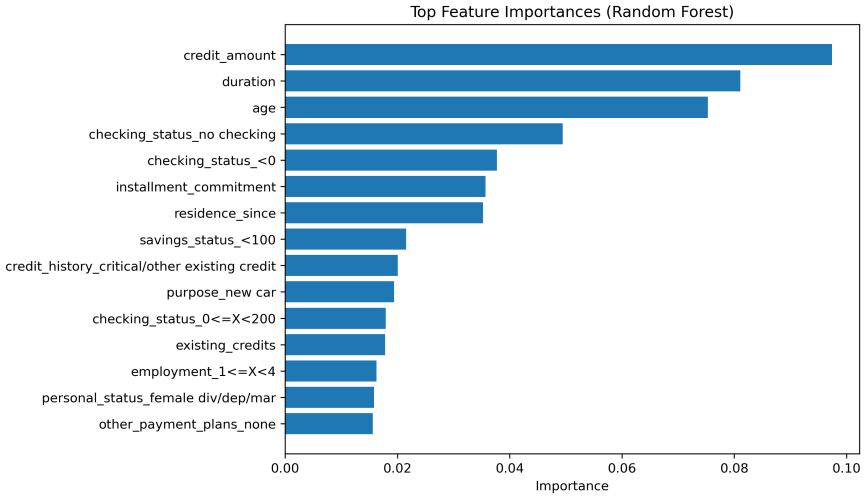

Figure 3: Feature importance from Random Forest

5 Discussion

Random Forest outperforms Logistic Regression in predictive accuracy, while Logistic Regression remains more interpretable. This reflects the common trade-off between performance and explainability in financial machine learning systems [4].

6 Conclusion

This study demonstrates a complete ML pipeline for credit risk prediction. Ensemble methods improve performance, while linear models provide interpretability.

Future work includes:

- Hyperparameter tuning
- Gradient boosting methods
- Explainable AI techniques

References

References

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